DNA Biometrics

Introduction to DNA Biometrics

- DNA (Deoxyribonucleic Acid) is the **genetic blueprint** of an individual.
- Present in almost every cell of the human body.
- DNA biometrics uses genetic variations for identification.
- Considered the **most accurate biometric trait**.

What is DNA?

- Double-helix molecular structure.
- Made of nucleotides:
  - Adenine (A)
  - Thymine (T)
  - Cytosine (C)
  - Guanine (G)
- Base pairing:
  - A–T
  - C–G

Structure of DNA

- Two antiparallel strands.
- Sugar-phosphate backbone.
- Base pairs form genetic code.
- Organized into chromosomes (23 pairs in humans).

Genetic Variation Used in Biometrics

- DNA identification is based on:
- Short Tandem Repeats (STRs)
- Variable Number Tandem Repeats (VNTRs)
- Single Nucleotide Polymorphisms (SNPs)
- Most common method: **STR profiling**

What are STRs?

- Short repeating DNA sequences (2–6 base pairs).
- Repeated multiple times at specific locations (loci).
- Number of repeats varies among individuals.
- Highly discriminative.

DNA Biometric System Architecture

- Sample Collection
- DNA Extraction
- DNA Quantification
- PCR Amplification
- STR Analysis
- Profile Generation
- Database Comparison
- Decision Making

Block Diagram

- Sample → Extraction → PCR Amplification → STR Detection → Digital Profile Creation → Database Matching → Identification

Sample Collection

- Common biological sources:
  - Blood
  - Saliva
  - Hair roots
  - Skin cells
  - Bone (in forensic cases)

Important: Avoid contamination

DNA Extraction

- Steps:
  - Cell lysis (breaking cell membrane)
  - Removal of proteins
    - Protease enzymes
    - Chemical reagents
  - Purification of DNA
    - Salts
    - Cellular contaminants
    - Enzymes
  - Isolation of genetic material
    - Pure DNA is collected
    - Stored for further analysis(PCR, STR profiling)

Methods

- Organic extraction
  - Uses organic solvents
  - Separates DNA based on solubility differences
- Chelex method
  - Uses Chelax resin to bind metal ions
  - Prevents DNA degradation
- Silica based extraction
  - DNA binds to silica membrane in presence of salt
  - Contaminants are washed away

DNA Quantification

Purpose:

- Measure DNA concentration.
- Ensure optimal amount for PCR

DNA Quantification ensures:

✓ Standardized input for STR profiling
✓ Reduced error rates
✓ Improved matching reliability
✓ Better statistical confidence

Techniques:

- Spectrophotometry
  - DNA absorbs UV light at 260nm
  - Concentration is calculated using absorbance.
  - 1 A_{260}=50 µg/mL of DNA
  - (ideal ≈ 1.8 for pure DNA)
- Fluorescent quantification
  - Uses DNA-binding fluorescent dyes.
  - Fluorescence intensity ∝ DNA concentration.
- Real-time PCR
  - Most advanced and accurate method.
  - Amplifies DNA while measuring fluorescence in real time.
  - Quantifies only amplifiable DNA

PCR(Polymerase Chain Reaction) Amplification

Purpose:

- Amplify specific STR regions.
- Produce millions of copies

- Process (3-step process)
  - Denaturation (95°C)
    - This "unzips" the DNA.
  - Annealing (50–65°C)
    - This marks the specific STR region to be copied
  - Extension (72°C)
    - effectively duplicating the targeted DNA segment.

- Repeated for 25–35 cycles.

STR Analysis

- Amplified DNA fragments separated by:
- Capillary electrophoresis
- Fragments labeled with fluorescent dyes.
- Detector measures fragment length
- Number of repeats determined.

DNA Profile Generation

Output

- Numeric representation of repeat counts.
- Example:

Locus	Allele 1	Allele 2
D8S1179	12	14
D21S11	29	30

- This forms the **DNA biometric template**.
  - The system produces a **numeric representation of repeat counts** at specific genetic locations called **loci**.
  - The number of times a short DNA sequence repeats at specific locations (STR loci).

Each person has: **Two alleles per locus**

One inherited from the mother

One inherited from the father

DNA Template

- Is a structured numeric profile
- Is used for comparison during identification
- Does NOT store complete genetic information
- Is compact and computationally efficient
- Template used in
  - Used in forensic identification
  - Paternity testing
  - Criminal investigations
  - Missing person identification

Matching Algorithm

- Comparison based on:
  - Allele matching across multiple loci.
    - Each person has **two alleles per STR locus**
    - A DNA profile contains repeat numbers at multiple loci (e.g., 13–20 loci in forensic systems).
    - If all loci match → strong evidence of identity.
  - Probability calculation.
    - What is the probability that this match occurred by chance?
      - Using allele frequency databases
      - Applying population genetics principles
      - Calculating locus-wise probabilities

- Random Match Probability (RMP).
  - The probability that a randomly selected unrelated individual from the population would have the same DNA profile.
  - Mathematically: $RMP = P1 \times P2 \times P3 \times ... \times Pn$
  - Pi = probability of genotype at locus I
  - n = number of loci
  - Because loci are statistically independent, probabilities are multiplied (Product Rule).

Decision Rule:

- If all STR loci match → Identity confirmed.
  - Full match at all tested loci
  - Very low RMP (e.g., 1 in 10^{12} or lower)
  - Statistical threshold satisfied

Statistical Model

DNA identification is not just visual comparison — it is based on **probability theory and statistics**.

Probability of match:

P(match) = Product of individual locus probabilities

Each STR locus has:

Certain allele frequencies in the population

A probability that a random person has that allele combination

Example

- If each locus probability = 1/10
  For 15 loci:
- $1/(10^{15})$ → Extremely low chance of random match

Advantages

- Highest accuracy
- Extremely reliable
- Cannot be easily forged
- Works even with partial samples
- Useful in criminal investigations

Disadvantages

- Expensive
- Time-consuming (hours to days)
- Privacy concerns
- Ethical issues
- Requires laboratory setup
- Not suitable for real-time authentication

Applications

- Forensic investigation
- Criminal identification
- Paternity testing
- Disaster victim identification
- Missing persons identification

Privacy & Ethical Concerns

- Contains sensitive genetic information.
- Risk of misuse
- Requires strict legal regulations
- Data protection laws necessary

Comparison with Other Biometrics

Biometric	Accuracy	Speed	Cost
Fingerprint	High	Fast	Low
Iris	Very High	Fast	Medium
Retina	Extremely High	Moderate	High
DNA	Highest	Slow	Very High